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Paleomagnetic evidence that the central block of Salinia (California) is not a far-traveled terrane

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Abstract. New paleomagnetic results from Late Cretaceous (75-85 m.y.) red beds on the central block of Salinia indicate that Salinia was located within 6° (in latitude) of its current cratonal North American position during the Late Cretaceous (after correction for Neogene San Andreas Fault transport). The red beds formed as alluvial-fan overbank deposits with hematite cement deposited directly on Salinian granites in the La Panza Range. Paleomagnetic analysis shows two components of magnetization in the red beds, a low-blocking-temperature present-day overprint residing in goethite and a high-blocking-temperature (>600°) component residing in hematite. The hematite magnetization is a chemical remanent magnetization which formed soon after deposition during pedogenesis. The bedding-corrected hematite remanence contains a magnetic polarity stratigraphy with antipodal normal and reversed directions. Twenty-three Class I sites (α_{95} < 20°) have an average hematite direction with inclination = 54.4° and declination = 18.2° (α_{95} = 6.1°) after structural correction. These paleomagnetic data suggest that Salinia resided at about 35°N latitude during the Late Cretaceous, within 6° of its current location adjacent to cratonal North America. By contrast, a summary of paleomagnetic data from the Peninsular Ranges terrane and the Sur-Obispo terrane, which are currently outboard of Salinia, shows northward transport of these terranes of 12° to 22° relative to their current locations in North America since the Cretaceous. The offsets increase systematically away from the craton with the most outboard Sur-Obispo terrane (which is composed of accretionary prism and distal forearc material) showing the largest degree of northward translation.

1. Introduction

Large-scale tectonic transport of crustal blocks (tectonostratigraphic terranes) northward along the west coast of North America has been proposed by many workers over the past 20 years [e.g., *Beck*, 1976, 1980; *Coney et al.*, 1980]. Such large-scale transport has been called for in part to accommodate paleomagnetic results which suggest that large parts of westernmost North America originated at anomalously low paleolatitudes compared to similar-aged rocks from cratonal North America. One of the most controversial regions of proposed northward transport has been the margin of coastal California west of the San Andreas fault. Coastal California is composed of several tectonostratigraphic terranes including (from north to south) the Point Arena terrane, Salinia, the Sur-Obispo (composite) terrane (SO; also known as the Nacimiento block), the Tujunga terrane, and the Peninsular Ranges (composite) terrane (PRT) (Figure 1).

Salinia, which is the best known of these terranes, is located west of the San Andreas fault and east of the Sur-Nacimiento fault zone. *Ross* [1978] divided Salinia (equal to his Salinian block) into four separate blocks (Figure 1), the largest of which, the central block, is the focus of this study. The boundaries of Salinia are not completely agreed upon, as some authors [e.g., *Mattinson*, 1990] consider *Ross's* [1978] southeastern block (Barrett Ridge block) to be part of Salinia, while others [e.g., *Page*, 1982] consider it to be more related to rocks of the neighboring San Gabriel Mountains (Figure 1, Tujunga terrane). Also, some authors include rocks around Point Arena as part of Salinia [e.g., *Mattinson*, 1990], while others [e.g., *Kanter and Debiche*, 1985] consider them to be the separate Point Arena terrane. Salinia is composed of Cretaceous granitic (continental arc) basement and framework metamorphic rocks overlain by Late Cretaceous and Paleogene marine (forearc basin) strata [*Ross*, 1978; *Page*, 1982; *Mattinson*, 1990]. The neighboring Tujunga terrane is composed of similar rocks but also includes older Mesozoic and Precambrian granites [*Anderson et al.*, 1992]. The Sur-Obispo composite terrane (SO), which is outboard of Salinia, occurs west of the Sur-Nacimiento fault and north of a poorly understood suture zone in the western Transverse Ranges [*Vedder et al.*, 1983]. It is made up of two terranes, the San Simeon and Stanley Mountain terranes [*Blake et al.*, 1982]. The San Simeon terrane consists of an accretionary wedge melange similar to the Franciscan Complex, while the Stanley Mountain terrane is composed of Jurassic ophiolite overlain by Jurassic, Cretaceous, and Paleogene marine (forearc basin) strata [*Vedder et al.*, 1983]. Several authors [e.g., *Vedder et al.*, 1983] have suggested that SO and Salinia amalgamated by the Late Cretaceous, so that their subsequent terrane histories can be considered together. The Point Arena terrane includes Jurassic and Cretaceous(?) spillites in fault contact with Late Cretaceous and Paleogene forearc basin strata. The Peninsular Ranges terrane (PRT), located in southern and Baja California (Figure 1), is a composite terrane composed primarily of

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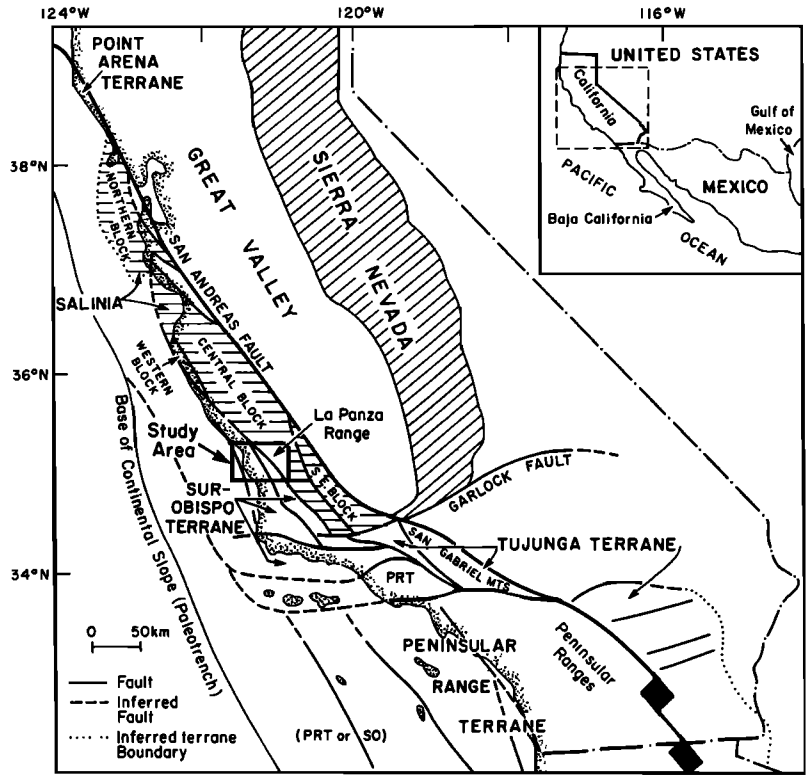


Figure 1. Locality map for terranes of coastal California. Terranes of coastal California that are discussed in this paper are, from north to south, the Point Arena terrane, Salinia, the Sur-Obispo (composite) terrane (SO), the Tujunga terrane, and the Peninsular Ranges (composite) terrane (PRT). The four blocks of Salinia (horizontal hachures) as described by *Ross* [1978] are labeled for clarity.

Cretaceous granitic basement and metamorphic framework rocks with some older Mesozoic granites, all of which are overlain by Cretaceous and Paleogene forearc basin strata [*Bottjer and Link*, 1984]. The boundaries of the PRT are the San Andreas and San Gabriel faults to the east and the Oak Ridge fault to the north [*Lund and Bottjer*, 1991].

The present-day paired configuration of Salinia (outboard) and the Sierra Nevada Cretaceous granitic batholith (inboard, Figure 1) has suggested to some workers that Salinia belonged at the southern end of the Sierra Nevada batholith, with the present configuration due to Neogene offset on the San Andreas fault [*Hamilton*, 1969; *Burchfiel and Davis*, 1972]. In the early 1980s, however, new paleomagnetic evidence was introduced which suggested that Salinia and SO were far-traveled terranes which originated much farther to the south with respect to cratonal North America and were displaced thousands of kilometers to the north and accreted to North America by Eocene time [*Champion et al.*, 1981, 1984; *McWilliams and Howell*, 1982]. Isotopic studies reported by *Ross* [1984] and *Silver and Mattinson* [1986], however, contradict the paleomagnetic data because they link the Cretaceous granitic rocks of the Salinian batholith with the Sierra Nevada batholith.

These two separate lines of evidence, paleomagnetic data and isotopic data, have led to two fundamentally different interpretations of the geologic history of coastal California, and thus far no new data have come to light to help resolve this dilemma. This paper presents new paleomagnetic data from Late Cretaceous (75-85 m.y.) strata on the central block of Salinia that links Salinia to the North American craton by Late Cretaceous time thus removing the need for Salinia to be a far-traveled terrane which has moved northward thousands of kilometers. Late Cretaceous red beds in the La Panza Range (Figure 1) have yielded a paleomagnetic vector direction with inclination = 54.4° and declination = 18.2° which we interpret to represent a primary magnetization of Late Cretaceous age. A paleolatitude estimate derived from these data indicates that Salinia was within 6° (500-700 km) of its current cratonal North America position in the Late Cretaceous.

2. Previous Paleomagnetic Results

Paleomagnetic studies of California coastal terranes were initiated 20 years ago by *Teissere and Beck* [1973], and since then, more than 40 studies have been carried out in these terranes. Also, more than 20 studies have been carried out on equivalent-aged rocks east of the San Andreas fault in California, which traditionally have been considered to be part of cratonal North America (for the last 120 m.y.). These various paleomagnetic studies have been recently summarized by *Champion et al.* [1986], *Lund and Bottjer* [1991], and *Lund et al.* [1991]. The paleomagnetic studies can be broadly lumped into three groups for discussion: California cratonal rocks located primarily east of the San Andreas fault, southern California rocks of the PRT, and coastal rocks related to SO. The central argument, which is the focus of this study, is

whether Salinia is tectonically more closely related to cratonal California or to SO.

Paleomagnetic studies have been carried out on several granite bodies of the Sierra Nevada batholith (Table 1). Paleolatitude estimates derived from these studies can be compared with their expected paleolatitudes derived from the North American apparent polar wander path (APWP) [*Irving and Irving*, 1982]. This difference is termed relative paleolatitude or $\Delta\lambda$ [e.g., *Lund and Bottjer*, 1991]; negative (positive) values imply that the studied rocks originated at a more southerly (northerly) latitude than that expected if they were part of cratonal North America. The results, listed in Table 1 and displayed in Figure 2a (open symbols older than 80 m.y.), indicate that the Cretaceous Sierra Nevada batholith formed at paleolatitudes not significantly different from those expected from the North American APWP. However, as was noted by *Frei et al.* [1984] and *Frei* [1986], the average paleolatitude of the Sierra Nevada batholith is 6°± 6° lower than expected from the North American APWP; this could indicate, but does not require, as much as 600 km of northward transport of the Sierra Nevada batholith and related rocks along the cratonal margin since Late Cretaceous time. Paleomagnetic studies of younger rocks from eastern California (including the Tujunga terrane which the geologic study of *Tosdal* [1990] indicates was part of cratonal California in the Late Cretaceous) are consistent with those of the Sierra Nevada batholith (Figure 2a). Individually, paleolatitude estimates from studies of the younger rocks are never significantly different from that expected from the North American APWP, but, as a group, the results are also biased to low relative paleolatitudes consistent with a few hundred kilometers of northward movement since Late Cretaceous time. On the basis of these data, we have derived a relative paleolatitude ($\Delta\lambda$) drift curve for cratonal California over the last 100 m.y. (Figure 3) and used this curve as a baseline for assessing whether terranes in coastal California should be considered far traveled (allochthonous).

A large number of paleomagnetic studies have been carried out on the PRT (equivalent to the Baja-Borderland Allochthon of *Champion et al.* [1986] and recently summarized by *Lund and Bottjer* [1991] and *Lund et al.* [1991]. The paleolatitude estimates from those studies are shown in Figure 2b. As has been noted before, the PRT paleolatitudes are consistently lower than equivalent-aged rocks of eastern California. (Paleolatitudes of all rock units which are discussed in this paper and located west of the San Andreas Fault have been estimated after correcting for Neogene structural disruption using the reconstruction model of *Butler et al.* [1991]; see Table 1.) A relative paleolatitude drift curve for the PRT, shown in Figure 3, illustrates the offset between the PRT and the Sierra Nevada batholith. We interpret these curves to indicate that the PRT was about 600-800 km farther to the south with respect to the Sierra Nevada batholith in the Late Cretaceous than it was at the beginning of Neogene motion on the San Andreas Fault.

Butler et al. [1991] have suggested, alternatively, that the interpretations of paleomagnetic data from the PRT and related rocks of coastal California (see below) are flawed. They argue that the paleomagnetic data derived from depositional/postdepositional remanent magnetizations (DRM/PDRMs) in sedimentary rocks (solid symbols in Figure 2) may have "inclination errors" associated with them which make their paleolatitude estimates appear anomalously low. However, we have noted elsewhere [e.g., *Lund et al.*, 1991] that paleomagnetic results from thermoremanent magnetizations (TRMs) and chemical remanent magnetizations (CRMs) (open symbols in Figure 2) of similar age and locality, which should not be subject to potential "inclination errors", are not significantly different from the sedimentary rock results (Figure 2). One would expect to see systematic paleolatitude offsets between the DRM/PDRM and TRM/CRM data within each of the three data sets in Figure 2 if significant "inclination errors" were common. *Butler et al.* [1991] have also called into question TRM results from the Cretaceous batholiths because they may have undergone tilting after the TRM was acquired, producing anomalously shallow inclinations and paleolatitudes. *Beck* [1992] has considered the likelihood of this type of error in general, and delineated methods for assessing its importance in any study. *Lund et al.* [1991] have argued that the TRMs in the Peninsular Ranges batholith are either posttilting or unaffected by regional tilting, for their remanences mimic those of PRT layered rock units, and all of the data show clear evidence for a systematic 30° rotation of the PRT that occurred during Paleogene time.

The remaining paleomagnetic data from rocks west of the San Andreas fault have usually been considered with the Sur-Obispo terrane (SO). *McWilliams and Howell* [1982] were the first to study rocks of SO. They analyzed Jurassic basalts and Cretaceous sedimentary rocks of the Stanley Mountain terrane. Their results (Figure 2c) suggested that SO was located at much lower paleolatitudes in Cretaceous time than expected if the rocks had developed in their present North American location. Similarly, *Kanter and Debiche* [1985] determined that rocks of the Point Arena terrane had also been located at similarly low paleolatitudes in Cretaceous and Paleogene time (Figures 2c and 3). (However, *Hagstrum and Murchey* [1996] suggest, based on new results from red cherts from the Coast Range ophiolite, that SO had a paleolatitude of 35° ± 4° in the Late Jurassic, suggesting that it has not moved far northward since deposition.) Groups studying both the PRT and SO have argued that their paleomagnetic results also applied to Salinia and used their paleomagnetic data to develop a terrane history for Salinia, SO, and the Point Arena terrane. However, until this study there have been no Mesozoic-aged paleomagnetic results from Salinia itself suitable for estimating its paleolatitude. The unresolved issue, which is addressed in this paper, is what is the paleolatitude of Salinia in the Cretaceous? Is its paleolatitude anomalously low like SO results or consistent with "cratonal" California like the Eocene data? Our new result from Late Cretaceous red beds of Salinia is discussed next.

3. New Paleomagnetic Results

3.1. Location and Stratigraphy

Unnamed Late Cretaceous sedimentary strata up to 3 km thick are extensively exposed on Salinia [e.g., *Howell et al.*, 1977; *Grove*, 1989, 1993]. In the region of Pozo Summit in the La Panza Range (Figures 4 and 5), interbedded nonmarine sandstones and disorganized boulder conglomerates directly

Table 1. Summary of Paleomagnetic Results From Terranes and the Adjacent North American Craton in California (United States) and Baja California (Mexico)

Site	Reference	Site λ*	Site ϕ*	Age, Ma	N	I	D	α_{95}	Conf†	Pλ(I)	I(NAM)	D(NAM)	α_{95}	Pλ(NAM)	Δλ	ROT (R)
							Peninsular Ranges Terrane									
							(PRT - Layered Rocks)									
CED	*Smith and Busby [1991]*	26.1	247.8	100±13	10/38	37	---	5.4	b	20.6±3.7	53.5	-25.0	10	34.0	-13.4±9.1	---
VF	*Hagstrum et al. [1985]*	25.8	248.1	92±20	3/17	41.4	9.6	15.5	b	23.8±11.9	53.7	-24.9	4	34.2	-10.4±12.8	34.5±17.2
CH2	*Champion et al. [1986]*	30.4	243.6	90±2	97/-	44	24	2.6	b	25.8±2	59.4	-25.6	4	40.3	-14.5±3.8	49.6±5.4
KSA	*Fry et al. [1985]*	31.6	245.3	83±9	20/80	45.1	-36.4	6.7	a,b	26.6±5.4	60.6	-24.5	7	41.6	-15.0±7.8	-11.9±11.0
KSM	*Morris et al. [1986]*	31.6	244.9	82±10	12/48	46.9	90.8	5.8	a,b	28.1±5.9	60.7	-24.9	7	41.7	-13.6±7.9	114.9±11
KPB	*Morris et al. [1986]*	27.7	247.0	76±9	17/68	44.5	3.5	8.2	a,b	26.2±6.5	56.2	-19.8	7	36.8	-10.6±8.6	23.3±11.9
FKR	*Filmer and Kirschvink [1989]*	29.2	246.3	75±5	131	44.6	5.3	2.5	a,b	26.2±2.0	57.7	-19.1	7	38.4	-12.2±5.9	24.4±8.2
SIL	unpublished	31.6	245.3	64±3	12/48	44.2	-28.1	7	a,b	25.9±5.5	57.0	-12.6	5	37.6	-11.7±6.9	-15.5±9.3
SSU	unpublished	31.3	245.0	59.5±2	12/44	43	86.4	9.9	a,b	25.0±7.7	55.3	-9.3	5	35.8	-10.8±8.9	95.7±11.7
FLN	*Flynn et al. [1989]*	26.6	248.7	55	44	47.6	-17	4.4	a	28.7±3.7	49.7	-8.5	4	30.5	-1.8±4.7	-8.5±6.5
LLJ	unpublished	31.3	245.0	51±2	10/39	47.6	69.1	8.5	a,b	28.7±7.3	55.0	-8.1	4	35.5	-6.8±7.5	77.2±10.7
LAJ	unpublished	30.5	245.8	49.5±2	30/115	45	8.4	3.3	a,b	26.6±2.6	54.1	-8.1	4	34.6	-8.0±4.1	16.5±5.1
POW	unpublished	30.5	245.8	42.5±1	12/47	47.5	-3.2	5.6	a	28.6±4.7	51.5	-7.8	4	32.2	-3.6±5.5	4.6±7.5
SES	unpublished	30.7	244.5	38	7/28	44.4	65	5.2	a	26.1±4.1	50.7	-6.5	4	31.4	-5.3±5.6	71.6±7.0
M/20	*Hagstrum et al. [1987]*	22.5	252.5	20±2.5	14	33.5	0.3	8	a	18.3±5.2	37.7	-1.7	3	21.1	-2.8±6.8	2.0±8.1
M/15	*Hagstrum et al. [1987]*	22.5	252.5	15±2.5	10	42.2	-0.3	9	a	24.4±6.9	37.4	-1.5	3	20.9	3.5±7.1	1.2±10.1
M/10	*Hagstrum et al. [1987]*	22.5	252.5	10±2.5	29	41.6	-3.6	6.2	a	23.9±4.6	37.1	-1.2	3	20.7	3.2±5.5	-2±7.1
M/5	*Hagstrum et al. [1987]*	22.5	252.5	5±2.5	8	45.8	-5.2	10.2	a	27.2±8.4	37.1	-1.2	3	20.7	6.5±8.5	-4.0±12.0
LEV1	*Levi[1988]*	34.4	241.5	1.7±1	28/152	47	0.4	5.2	a,b	28.2±4.3	52.2	-1.5	3	32.8	-4.6±4.8	1.9±6.7
LEV2	*Levi[1987]*	34.3	241.5	1.7±1	33/198	48.1	1.8	4.4	a,b	29.1±3.8	52.1	-1.5	3	32.7	-3.6±4.3	3.3±6.0
LEV3	*Levi and Yeats [1984]*	34.4	241.5	1.5±1	36/46	55	---	5	a,b	35.5±5.1	52.2	-1.5	3	32.8	2.7±4.7	1.5±7.5
							(PRT - Batholith Rocks)									
SCB	*Teisserre and Beck [1973]*	31.0	246.0	110±10	18/100	49.5	3	5	-	30.3±4.4	58.1	-23.7	6	39.3	-9.0±6.2	26.7±8.1
SMG	*Hagstrum et al. [1985]*	30.8	245.9	110±3	14/139	48.1	9.3	3.8	-	29.1±3.2	58.5	-23.5	6	39.2	-9.9±5.7	33.3±6.9
PRB1	*Hagstrum et al. [1985]*	27.5	248.3	92±20	4/22	54	5.2	4.6	-	34.5±4.5	55.0	-25.6	6	35.8	-1.3±6.0	30.8±8.1
							Sur-Obispo/Point Arena Terranes									
PAS	*Kanter and Debiche [1985]*	36.2	240.2	120±40	---	10.1	---	3.4	-	5.1±1.7	65.4	-25.6	19	47.5	-42.4 ±15.4	----
BAJ	*Hagstrom and Sedlock [1990]*	26.1	247.6	105±15	16/102	23.4	31.4	9.3	b	12.2±5.3	53.7	-23.8	10	34.2	-22.0±10.9	55.4±12.9
FIG	*McWilliams and Howell [1982]*	33.0	243.9	91±7	10/60	12	23	11	c	6.1±5.7	61.4	-26.8	4	42.5	-36.4±9.3	49.8±10.1
PP	*Champion et al. [1984]*	34.7	241.2	72±6	4/128	37.8	---	5.3	a	21.2±3.7	62.6	-18.5	7	44.0	-22.8±7.0	----
PSP	*Champion et al. [1984]*	35.9	240.6	61±3	5/102	42.3	---	4.5	a	24.5±3.4	60.5	-10.6	5	42.9	-15.5±5.4	----
GER	*Kanter and Debiche [1985]*	36.2	240.2	51±13	---	43	---	2.8	-	25.0±2.2	60.0	-8.2	4	40.9	-15.9±3.9	----
SES2	*Liddicoat [1990]*	30.7	244.5	30±5	8/48	38.9	80.1	5.4	a,b	22.0±3.8	49.8	-4.9	3	30.6	-8.6±9.9	85.0±6.3
IB	*Kanter and Debiche [1985]*	36.2	240.2	23±2	---	48.3	---	14.8	-	29.3±13.1	54.8	-2.3	3	35.3	-6.0±12.1	----
HF1	*Hornafius [1985]*	31.5	244.7	15±1	82	44.8	91.7	4.9	b	26.4±3.9	49.2	-1.6	3	30.1	-3.7±4.6	93.3±6.2
HF2	*Hornafius [1985]*	31.5	244.7	14.5±0.5	109	48.4	79.6	4.6	b	29.4±3.9	49.2	-1.6	3	30.1	-0.7±4.4	81.2±6.2
HF4	*Hornafius [1985]*	31.5	244.7	12.5±1.5	41	48.8	66.9	4.8	b	29.7±4.2	49.2	-1.6	3	30.1	-0.4±4.5	68.5±6.5
HF5	*Hornafius [1985]*	31.5	244.7	9.5±1.5	58	47.7	35.9	4.9	b	28.8±4.2	49.0	-1.6	3	29.9	-1.1±4.6	37.5±6.5
							Salinia									
BUT	*Kanter [1988]*	35.7	240.7	49±6	30	54.1	---	4.9	a	34.6±4.9	59.5	-8.3	4	40.3	-5.7±4.7	----
							East California (plus Tujunga)									
							(East California - Layered Rocks)									
SFR	unpublished	33.4	244.9	65±5	7/21	53.8	25	6	a,b	34.3±5.9	58.9	-13.5	4	39.6	-5.3±5.8	33.2±9.1
TEJ	unpublished	34.9	240.7	50±5	7/21	56.1	-15	7.8	a,b	36.6±8.1	58.8	-8.2	4	39.5	-2.9±7.0	-6.8±12.0

Table 1. (continued)

Site	Reference	Site λ*	Site φ*	Age, Ma	N	I	D	α_{95}	Conf†	Pλ(I)	I(NAM)	D(NAM)	α_{95}	Pλ(NAM)	Δλ	ROT (R)
					(East California Layered Rocks) (continued)											
VAS	*Terres and Luyendyk [1985]*	32.3	245.0	23±6	31	54.2	---	6.7	a,b	34.7±6.7	50.7	-2.1	3	31.4	3.3±5.9	---
MIO	*Kanter and McWilliams [1982]*	35.2	241.8	20±5	12/73	63.7	-6.3	5.9	a	45.3±7.5	53.6	-2.1	3	34.1	11.2±5.3	-4.2±11.1
MOJ1	*Golombeck and Brown [1988]*	35	242.3	20±3	18/162	51.6	15.6	8.2	a,b	32.2±7.7	53.4	-2.1	3	33.9	-1.7±7.0	17.7±11.0
WAZ	*Calderone et al. [1990]*	34	246	16±4	65	49.8	-3.6	4.4	a	30.6±3.9	52.2	-2	3	32.8	-2.2±4.3	-1.6±6.2
BAR	*McFadden et al. [1990]*	35	243	15±2	125	47.5	-12.9	5.4	a,b	28.6±4.6	52.8	-1.5	3	33.4	-4.8±4.9	-11.4±7.0
ETR	*Carter et al. [1987]*	33.9	244.5	12.5±3	27	50.4	39.7	6.4	a	31.4±5.8	51.6	-1.4	3	32.2	-1.1±5.7	41.1±8.5
KON	*Kono et al. [1967]*	36.5	254.3	4±1	18	51.1	2.2	6.3	a	31.8±5.8	54.2	-1.1	3	34.7	-2.9±5.6	3.3±8.6
						(Sierra Nevada Batholith)										
BUC	*Gromme et al. [1967]*	39.9	238.7	120±20	9/126	72.1	-42.9	7.9	a	57.1±12.3	68.2	-27.2	19	51.3	5.8±16.5	-15.6±32.9
FRE	*Frei [1986]*	37.6	241.2	98±5	9/51	61.5	-27.7	3.7	c	42.6±4.4	64.9	-28.7	10	46.9	-4.2±8.5	1.0±13.3
GM	*Frei et al. [1984]*	38.2	240.4	90±5	14/114	66.1	-19	3.4	a	48.5±4.6	66.1	-28.6	4	48.5	0.0±4.2	9.6—8.3
MG1	*Frei et al. [1984]*	38.1	240.3	90±5	19/150	60.3	-27.7	3.3	c	41.2±3.8	66.1	-28.6	4	48.5	-7.2±4.1	0.9±7.2
MG2	*Frei et al. [1984]*	38.2	240.3	90±5	7/53	58.4	-18.8	3.8	c	39.1±4.2	66.1	-28.6	4	48.5	-9.3±4.4	9.8±7.6
GR	*Gromme and Merrill [1965]*	38.4	240.1	85±10	14/100	68.1	-24	9.6	c	51.2±13.6	67.4	-26.2	7	50.2	1.0±9.5	2.2±23.0
BVS1	*Kanter and McWilliams [1982]*	35.2	241.5	83±6	8/34	55.7	22.3	5.6	a	36.2±5.8	64.9	-25	7	46.9	-10.6±7.2	47.3±11.4
BVS2	*McWilliams and Li [1985]*	35.5	241.2	83±3	16/69	68.1	-23.5	3	c	51.2±4.2	65.1	-25.1	7	47.1	4.1±6.1	1.6±10.5
BVS3	*McWilliams and Li [1985]*	35	241.2	82±5	10/50	56	34.5	8	c	36.5±8.3	64.8	-24.8	7	46.7	-10.2±8.5	59.3±14.1

N,site/sample numbers; I, inclination; D, declination; PA, paleolatitude; NAM, cratonal North America; and ROT, rotation. Dashes indicate no information is available from other published studies.
* Site latitude and longitude after correction for Neogene structural disruption using the reconstruction model of *Butler et al.* [1991].
† Data confidence (Conf) is defined by (a) presence of both normal and reversed polarity and (usually) passage of a reversal test, (b) passage of a fold test, (c) locality mean directions (in situ) which are significantly different from present axial-dipole direction.

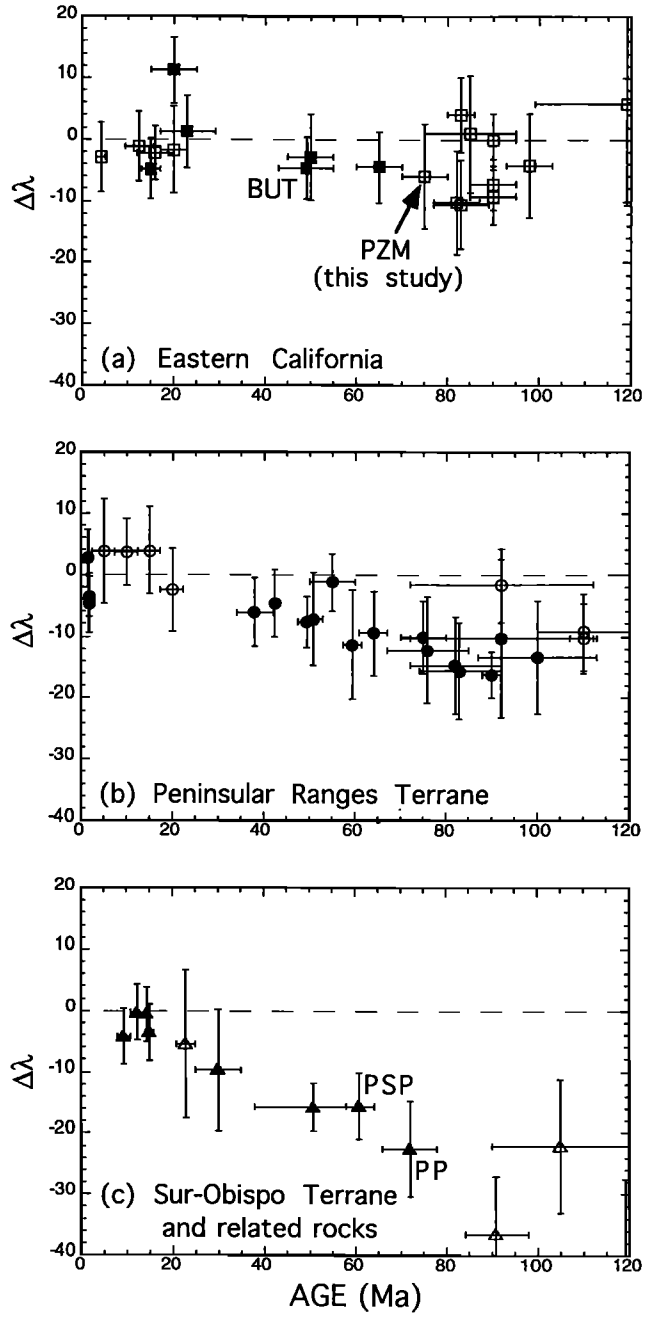


Figure 2. A summary of Cretaceous-Holocene paleomagnetic results from California. (a) Results from eastern (cratonal) California (squares). (b) Results from the Peninsular Ranges terrane (circles). (c) Results from the Sur-Obispo terrane and related rocks (triangles). Results west of the San Andreas Fault have been corrected for known Neogene transport using the reconstruction of *Butler et al.* [1991] (see also Table 1). Open symbols are paleomagnetic results derived from thermoremanent (TRM) and chemical (CRM) remanent magnetizations. Solid symbols are results derived from depositional or postdepositional (DRM/PDRM) remanent magnetizations. Paleomagnetic results from Salinia (Butano Sandstone (BUT), Pigeon Point Formation (PP), Point San Pedro (PSP), and the new result of this study, Pozo Mountain (PZM)) are labeled separately. See the text for a more thorough discussion of the data. (Modified from *Lund et al.* [1991].)

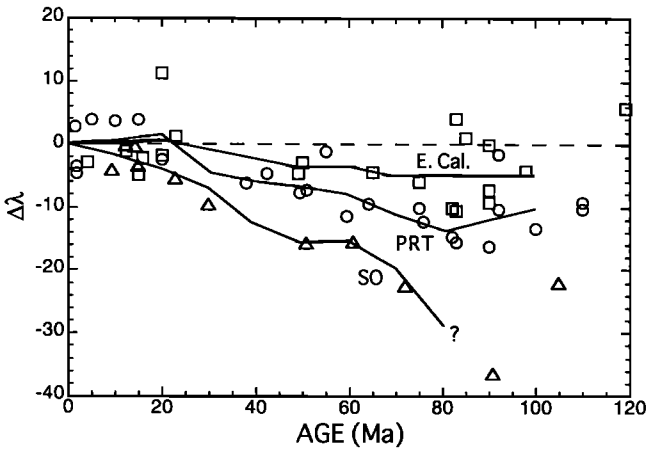


Figure 3. Relative paleolatitude drift curves for selected regions of California. The 20-Ma moving-average paleolatitudes for eastern California, the PRT, and SO are plotted together with the individual paleomagnetic results for each region. The tendency for the three paleolatitude curves to converge is attributed to active plate-margin dextral shear and northward transport of coastal terranes; the largest amount of transport occurred in the most coastal terranes, and the degree of transport decreased going inboard toward cratonal North America. Note that paleolatitude estimates for regions west of the San Andreas Fault have been corrected for known Neogene transport using the reconstruction of *Butler et al.* [1991]. (See also Figure 2 and Table 1.) (Modified from *Lund et al.* [1991].)

onlap Cretaceous granites of the central block of Salinia [*Howell et al.*, 1977]. Poorly sorted cobble to boulder conglomerates, interpreted as representing debris flow deposition, are interbedded with better-sorted cobble conglomerates and red beds interpreted as braided stream and overbank deposition [*Grove*, 1993]. *Grove* (1993) interpreted the depositional environment of these units as alluvial fan within a larger transgressive fan-delta system. The age of these units is likely Latest Cretaceous (75-85 m.y.); the ammonite Baculites occidentalis, which occurred from Late Campanian to Early Maastrichtian time, was found ~1 km below the top of the section [*Vedder*, 1977]. The nonmarine units are, in turn, overlain by shallow-marine sandstones and deep-sea fan deposits (Figure 6). Farther to the west around Santa Margarita Lake (Figure 4) and Nacimiento Lake, the basal sedimentary rocks onlapping Salinian granites are shallow marine sandstones and conglomerates, which are, in turn, overlain by deep-sea fan deposits [*Howell et al.*, 1977; *Grove*, 1989, 1993].

Paleomagnetic sampling was carried out in a variety of lithologies: shallow marine sandstones, deep-sea fan siltstones and mudstones, nonmarine sandstones, and nonmarine red beds. The marine sedimentary rocks (23 sites) were sampled near Santa Margarita Lake and along River Road (Figure 4), while the nonmarine sandstones and red beds (37 sites) were sampled along Pozo Road (Figures 4 and 5). It became apparent very early in this study that two independent research groups at the University of Southern California (USC) (K. Whidden, S. Lund, and D. Bottjer) and the U. S. Geological Survey (USGS) (D. Champion and D. Howell) were both studying the redbeds. We agreed to interactively carry on with both studies in parallel, including all laboratory analyses, and copublish our results.

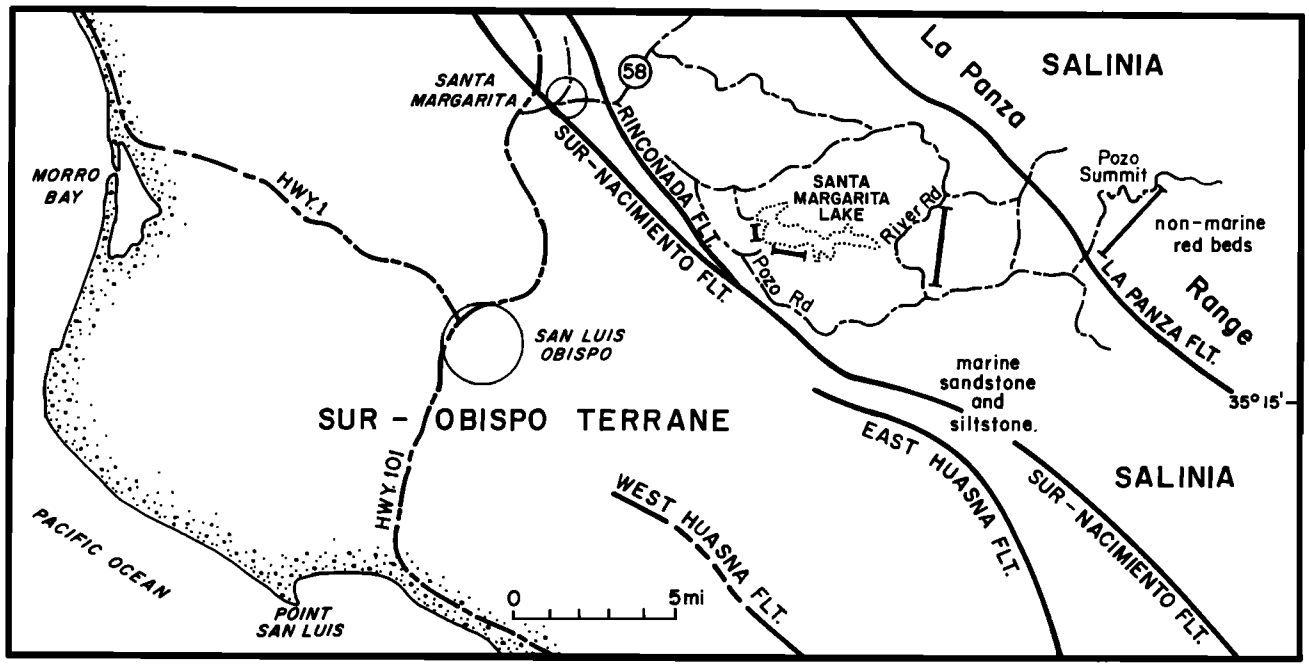


Figure 4. Locality map of the paleomagnetic study area in the La Panza Range (Figure 1). General sampling areas around Santa Margarita Lake and Pozo Summit are noted by bracketed lines. The Pozo Summit geology and sampling localities are also shown in more detail in Figure 5.

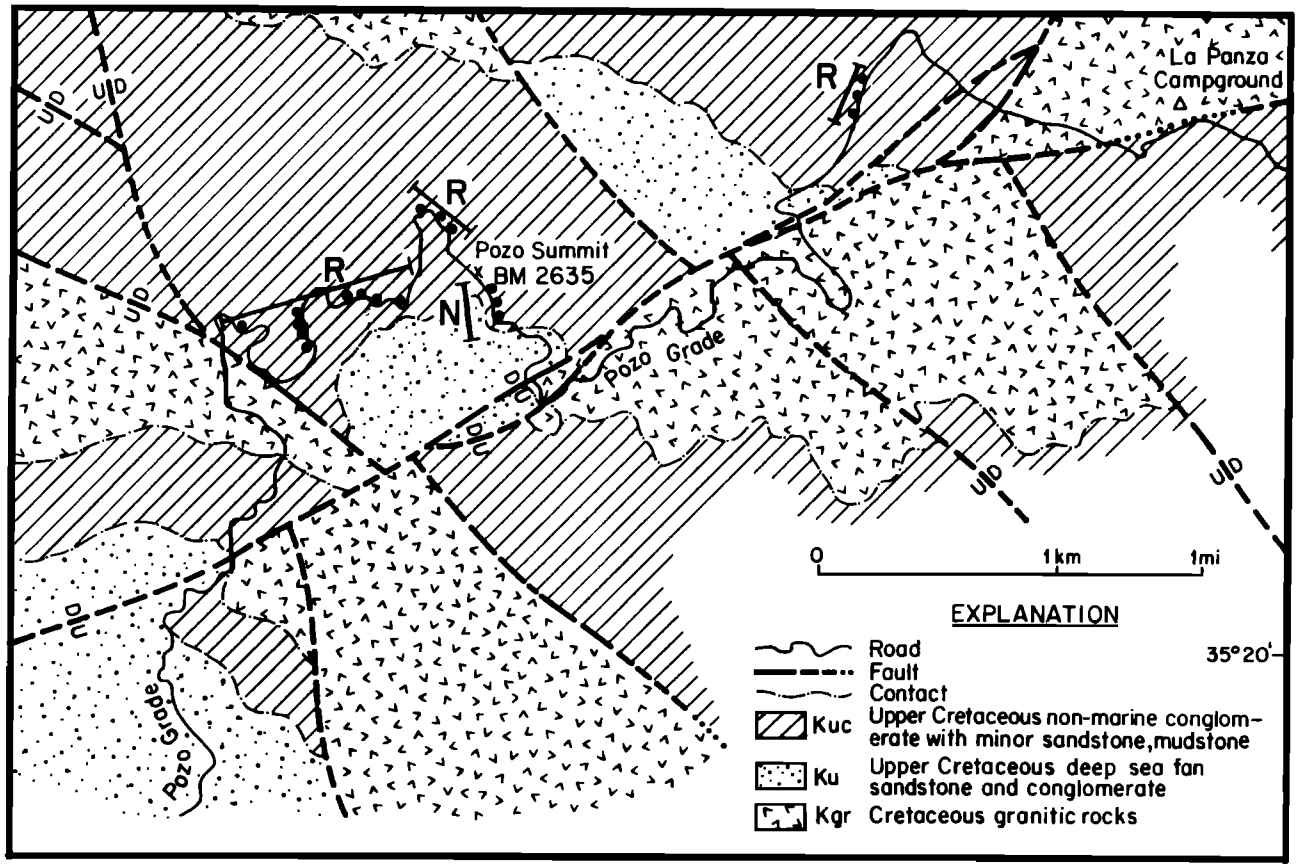


Figure 5. Map of Late Cretaceous granites and onlapping nonmarine and marine strata at Pozo Summit. The paleomagnetic sampling localities are shown by solid circles, and their interpreted magnetic polarity is noted. (Modified from *Howell et al.* [1977].)

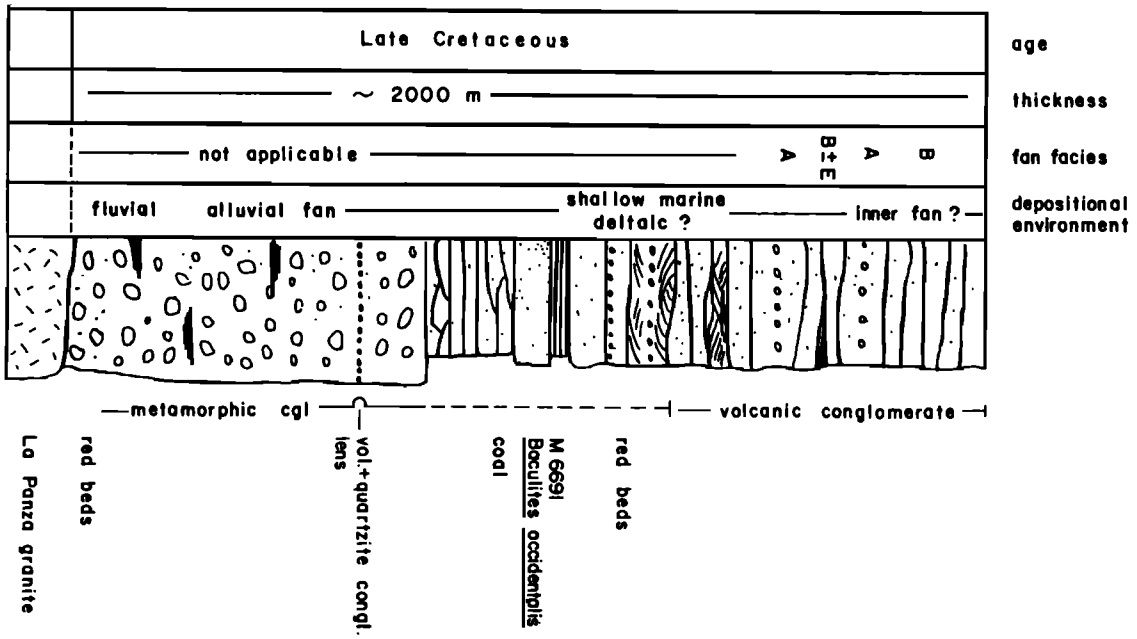


Figure 6. Generalized stratigraphic column of Late Cretaceous strata in the Pozo Summit region. Sampled red beds occur in the lower nonmarine strata. (Modified from *Howell et al.* [1977].)

3.2. Paleomagnetic Results

Paleomagnetic studies were carried out at both USC and the USGS in Menlo Park. At USC, the natural remanent magnetizations (NRMs) of all samples were initially measured and then the samples were all thermally demagnetized at 50°C steps up to 550°C, then at 25° steps to 675°C (for a total of 15 steps); at the USGS, similar procedures were used but the samples were thermally demagnetized at 20 different steps with step intervals decreasing from 50°C at low temperatures to about 10°C at high temperatures. Typical demagnetization diagrams for marine lithologies are shown in Figure 7, while diagrams for the nonmarine red beds are shown in Figure 8.

The marine lithologies (23 sites of sandstone, siltstone, and mudstone) all have relatively low-intensity NRMs which are dominated by a low-blocking-temperature (<300°C) characteristic remanence (ChRM) (Figure 7) with in situ normal-polarity field directions not significantly different from the present-day field. We interpret this NRM component to be a posttilting (recent) overprint that resides in authigenic goethite. There is intermittent evidence for a reversed component of magnetization at higher blocking temperatures, but the remaining remanences are too weak to characterize it. *Grove* [1989] noted the same pattern of behavior in paleomagnetic studies of similar marine rocks from around Nacimiento Lake (~ 30 km northwest of our study area) on the central block of Salinia.

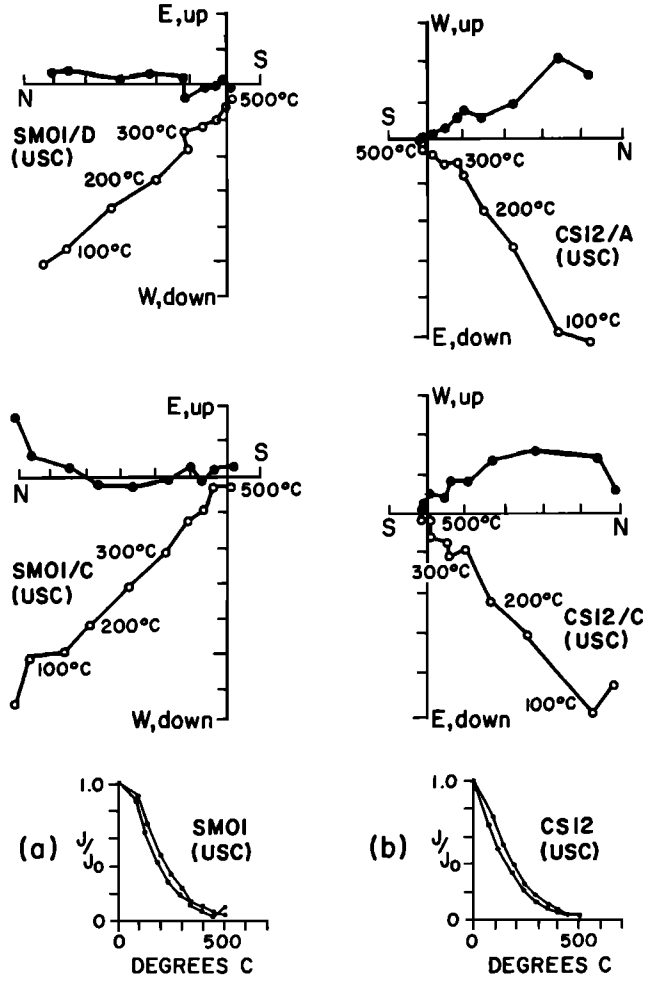


Figure 7. Representative zijderveld diagrams and blocking-temperature spectra for marine sandstones. Solid (open) symbols indicate horizontal (vertical) components of the natural remanent magnetization at each heating step. Note that most of the intensity is lost between 200° and 300°C. All data are plotted according to in situ coordinates.

The nonmarine lithologies (37 sites of gray to buff-colored sandstones and red beds) have more variable NRMs. Several of the gray to buff-colored sandstones have weak NRMs and behave like the marine rocks under thermal demagnetization. A few of the gray sandstones and all of the red beds, however, have much stronger NRMs (up to 2 orders of magnitude stronger than the marine rocks) and contain two ChRMs with distinctly different blocking-temperature spectra (Figure 8). All of the gray sandstones and some of the red beds have the same present-day goethite overprint seen in the marine rocks; the gray sandstones have the largest proportion of the goethite overprint, while the red beds have relatively less to almost none of this component (Figure 8). The nonmarine rocks also have a second ChRM with high blocking temperatures, mostly located above 600°C, which we interpret to reside in hematite. In a later section, we will discuss a variety of geological and rock magnetic evidence which suggests that the hematite is a chemical remanent magnetization (CRM) formed soon after deposition of the nonmarine rocks. We see almost no evidence for the presence of other magnetic phases, especially magnetite, in these rocks. (This is important to note in light of the fact that magnetite is the only observable magnetic phase which we have been able to identify in the underlying granites at Pozo Summit).

Vector directions associated with the high-blocking-temperature (hematite) ChRM in the nonmarine rocks were separated out for further analysis in one of two ways. The USGS lab used linear-line fits of paleomagnetic directions with well-determined linear trends of several demagnetization steps to estimate ChRMs. The USC lab used vector differences within well-determined linear segments of several demagnetization steps to estimate ChRMs. Several USC sites were subsequently analyzed using line fitting, and no significant differences were found in the overall vector directions. The vector directions were divided into three classes on the basis of size of vector α_{95} and linearity of the ChRMs. Class I sites (n=23) have α_{95}s of less than 20° and well-defined linear ChRMs, while class II sites (n=3) have α_{95}s larger than 20° and/or less well-defined ChRMs. In a few class I and class II sites, one sample ChRM direction was significantly different from the other samples and was discarded from the final site vector average. Several nonmarine sites (n=11) displaying no systematic ChRM were labeled as class III and discarded from further analysis. In all cases, the nonmarine class III sites were not markedly red in coloration and had been sampled to assess the variability in NRM within the overall nonmarine stratigraphic interval.

Table 2 lists the high-blocking-temperature ChRM vector results from the 26 class I and class II sites. Field corrected data (corrected for tilt of sediment bedding only) from the 23 class I sites have an average direction of I=21.0°, D=5.2°, and

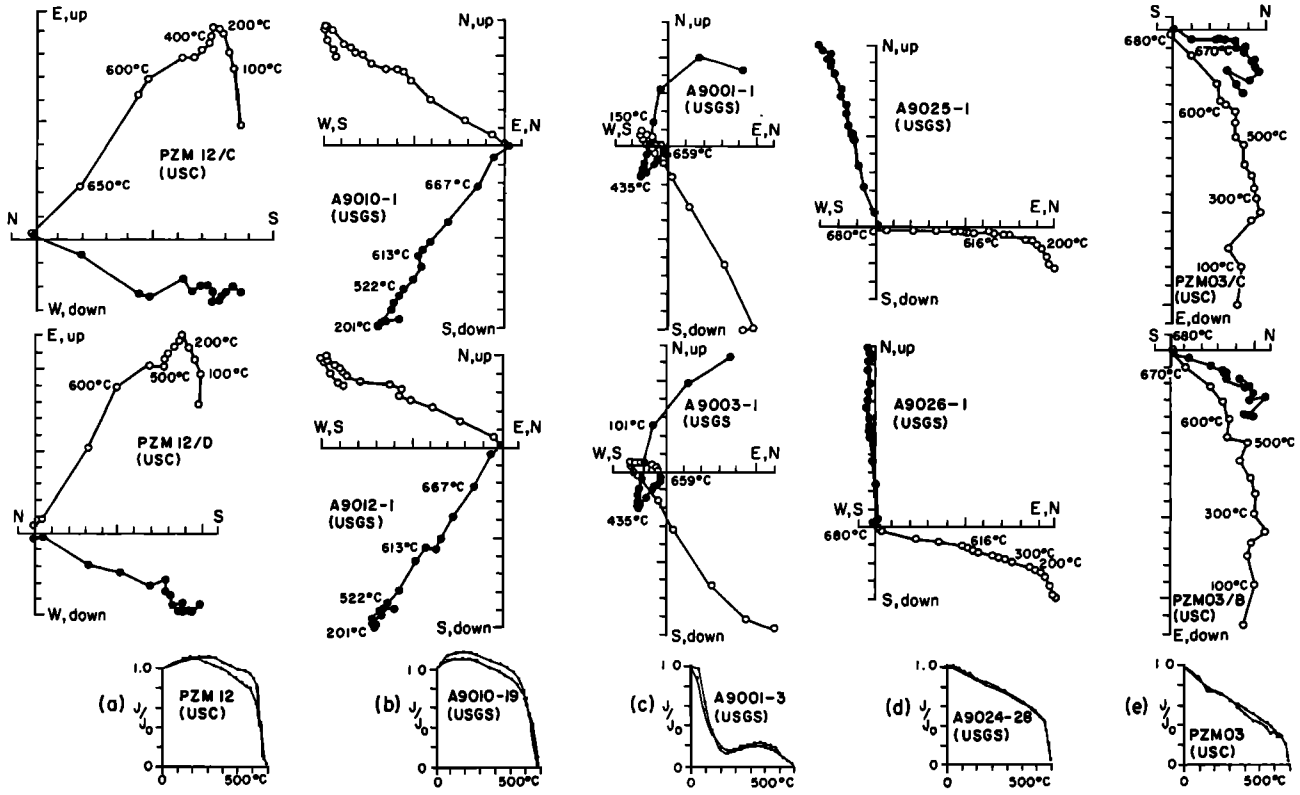


Figure 8. Representative zijderveld diagrams and blocking-temperature spectra for nonmarine red beds. Solid (open) symbols indicate horizontal (vertical) components of the natural remanent magnetization at each heating step. Note that most of the intensity is lost above 600°C. All data are plotted according to in situ coordinates.

α_{95}=10.2°; the bedding corrected data have an average direction of I=54.4°, D=18.2°, α_{95}=6.1° (Figure 9). On the basis of the lower α_{95} for the bedding-corrected data, we might argue that the data pass a paleomagnetic fold test. However, we consider this to be premature because the rocks around Pozo Summit (Figure 5) are highly faulted and there is no clear-cut spatial pattern to the structural corrections. A more conclusive paleomagnetic test is the fact that we have both normal- and reversed-polarity sites and that the sites are grouped spatially by polarity along Pozo Road (Figure 5). Unfortunately, again because of the complicated pattern of faulting, the exact stratigraphic relationships between the normal- and reversed-polarity sites cannot be determined. However, the normal-polarity sites all appear to occur up section from the reversed-polarity sites and closer to the boundary with the overlying marine strata. The normal- and reversed-polarity data also pass a reversal test (Figure 9) with an average normal-polarity direction of I=57.1°, D=9.2°, and α_{95}=9.8°, and an average reversed-polarity direction of I=-53.3°, D=-158.8°, and α_{95}=7.6°. Using the methodology of *McFadden and McElhinny* [1990], the data yield a positive reversal test with a "C" level of confidence.

3.3. Related Rock Magnetic Results

Before and during NRM demagnetization, magnetic susceptibility measurements were made to look for authigenic magnetic phases that might form in the heating process. No significant temperature-dependent changes in susceptibility were noted. Selected samples of the underlying granite at Pozo Summit were also collected to determine the likely source of detrital magnetic minerals. Thermomagnetic curves on two granite samples showed the presence of magnetite but no hematite.

4. Origin of the Hematite Remanence

4.1. Sedimentologic Observations

The nonmarine units, including the red beds, were deposited within an alluvial-fan environment [*Vedder et al.*, 1977; *Grove*, 1989, 1993]. The red beds occur overlying the granite and are intercalated with a massive disorganized boulder conglomerate (Figure 10) that has been interpreted as debris-flow deposits, with the red beds representing overbank deposits [*Vedder et al.*, 1977; *Grove*, 1989, 1993]. *Turner* [1980] described red beds developed in overbank environments as interbedded muddy layers and ripple-bedded and cross-bedded sandy layers; the mud horizons may show evidence of subaerial exposure such as mud cracks and often have abundant organic matter from plant growth; mottling can be extensive due to soil development.

The Pozo red beds range from very fine to medium-grained sandstones. Approximately half of our samples are uniform

TABLE 2. Summary of Paleomagnetic Results From the Pozo Red Beds

Site	N	In Situ		Bedding Corrected		α_{95}	ChRM	Polarity	Class
		I, deg	D, deg	I, deg	D, deg				
PZA-01	4/4	-70.1	143.8	-71.4	148.2	8.2	VDF	R	I
PZA-02	4/5	-17.4	-165.2	-59.2	-168.0	9.5	SEP	R	I
PZA-03	3/4	-9.5	-153.7	-40.8	-140.1	15.1	VDF	R	I
PZA-05	5/5	-24.1	178.6	-53.5	-170.1	7.0	SEP	R	I
PZA-07	3/4	-0.5	160.8	-47.1	-155.2	13.9	SEP	R	I
PZM-02	2/4	27.3	-24.7	56.9	-28.8	24.2	SEP	N	II
PZM-03	4/4	23.6	13.8	51.9	26.2	6.9	SEP	N	I
PZM-05	4/4	6.8	-4.9	60.7	10.7	6.7	SEP	N	I
PZM-06	3/4	19.2	-15.2	52.3	-0.5	18.9	SEP	N	I
PZM-09	4/4	2.7	143.6	-49.5	148.3	7.4	VDF	R	I
PZM-10	5/5	10.6	14.0	-42.2	141.6	26.8	SEP	R	II
PZM-11	3/3	10.1	164.4	-45.4	147.8	37.2	SEP	R	II
PZM-12	4/4	-21.3	-174.5	-55.4	-156.0	7.9	SEP	R	I
PZM-13	2/3	-8.5	-167.4	-48.5	-164.7	13.5	SEP	R	I
PZM-14	4/4	4.9	-148.3	-40.1	-142.3	8.8	SEP	R	I
CS-15	2/2	-12.5	-160.0	-54.4	-157.6	8.2	SEP	R	I
A9001	3/3	-17.7	-143.1	-37.4	-131.0	19.3	LLF	R	I
A9004	4/4	-22.3	-142.5	-45.8	-136.2	14.3	LLF	R	I
A9010	7/7	-27.6	-140.3	-46.0	-134.4	3.2	LLF	R	I
A9021	3/3	31.4	-1.0	61.6	21.9	7.9	LLF	N	I
A9024	4/5	13.6	0.1	45.1	10.5	4.6	LLF	N	I
A9029	2/3	22.3	-24.2	66.2	-21.9	0.9	LLF	N	I
A9032	4/4	-33.9	-176.0	-53.7	-162.4	5.5	LLF	R	I
A9039	2/3	-32.3	-175.2	-45.8	-167.9	2.0	LLF	R	I
A9046	3/4	-48.8	-170.3	-64.8	-146.0	12.2	LLF	R	I
A9052	9/15	-1.3	157.4	-52.8	162.3	5.6	LLF	R	I
Average directions									
All sites	26			52.6	11.9	9.6			
Class I sites	23			54.4	18.2	6.1			
Class I-N	6			57.1	9.2	9.8			
Class I-R	17			-53.3	-158.8	7.6			

N, number of samples used/collected; ChRM, characteristic remanence; VDF, vector difference; SEP, stable end point; LLF, linear line fit; N, normal; R, reversed.

red in color, while the rest exhibit wavy laminations or variegated coloration. The wavy laminations appear to represent thin layers of organic material, while the variegated color probably represents gleization (anaerobic decomposition of organic material and reduction of iron hydroxides and oxides [*Retallack*, 1991]) during soil development. The presence of root casts, coal seams, and leached horizons (Figure 10) all suggest that the red beds are

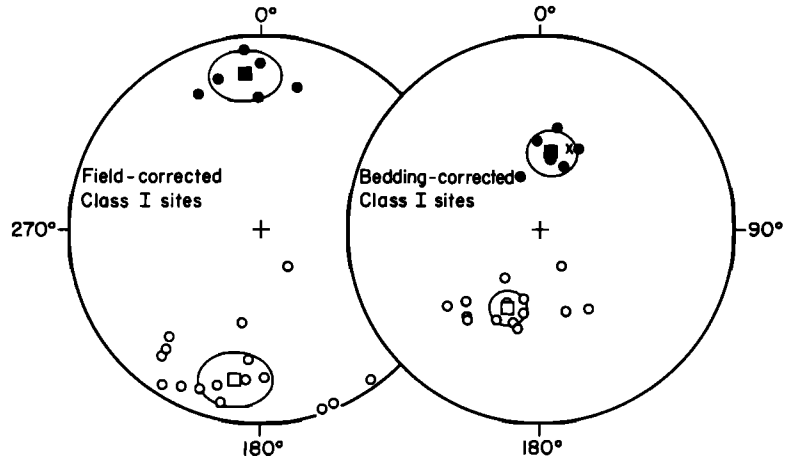


Figure 9. Stereonet plot of field- and bedding- corrected paleomagnetic data from class I sites showing antipodal nature of the normal- and reversed-polarity sites.





Figure 10. Field photographs of sampled red beds. Lighter regions are interpreted as leached horizons. Note variegated color patterns. Bedding follows contact between conglomerate and sandstones in both photographs.

paleosols, that is, sediments that were subaerially exposed and weathered following deposition. The leaching may have occurred during soil development (i.e., horizon E) or soon after burial because of local gleization.

In thin section, it can be seen that the main detrital constituents of the red beds are quartz (both monocrystalline and polycrystalline), feldspar (predominantly plagioclase with some potassium feldspar), mica (mainly biotite and some muscovite), and metamorphic rock fragments (Figure 11). The metamorphic rock fragments include polycrystalline quartz with sutured grains and schist fragments. The feldspars are commonly seriticized, with clay intergrowths. Biotite has also been altered, with clays and sometimes hematite developed in cleavage planes. Muscovite is less commonly altered than biotite but shows similar forms of alteration. A few samples show evidence of chlorite. There are also rare opaque detrital grains of magnetite or, less commonly, hematite. The red beds are generally poorly to moderately sorted, with a polymodal grain size distribution. All samples are grain-supported, with subangular to subrounded grains. Overall the sandstones are texturally immature.

The primary cement is hematite. It may occur as a relatively uniform cement surrounding detrital grains (Figure 11) or as irregular patches of more and less hematite cement (Figure 12). It is often intergrown with finely disseminated clays (Figure 11) and occasionally with chlorite. The clays probably

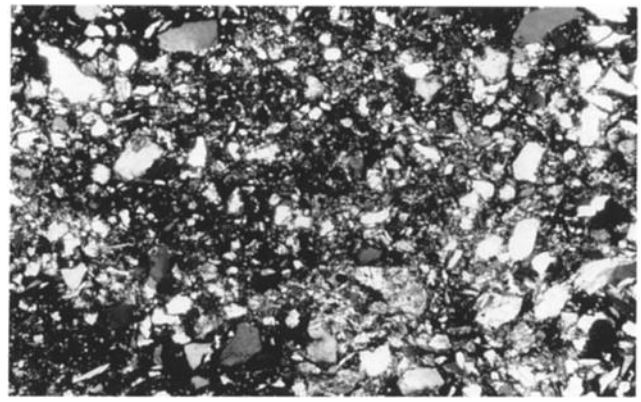


Figure 11. Thin section photomicrograph (polarized light) showing detrital components of red beds with well-developed hematite cement (dark material between grains). Photomicrograph is 3.5 mm across.

represent pedogenic material that has coated the grains during soil development (Figure 13). In mottled samples, the white or greenish regions are dominated by pedogenic clays, with some hematite patches occurring as partial grain coatings. Many samples have patches of quartz overgrowth, which likely formed during late burial of the strata because of intergranular pressure solution or dewatering and illitization of smectitic clays [*Retallack,* 1988, and references therein].

4.2. Evidence for Primary CRM in Pozo Red Beds

The primary remanance in the Pozo red beds is interpreted to be a chemical remanent magnetization (CRM), rather than a detrital remanent magnetization (DRM), because most of the hematite seen in the red beds is in the form of a chemically precipitated cement and not detrital grains. In addition, thermomagnetic curves of the magnetic minerals in the underlying granite, which was the source for red bed sediments, show that magnetite is the main magnetic mineral. The blocking-temperature spectra indicate that magnetite contributes at most a small percentage of the remanance in a

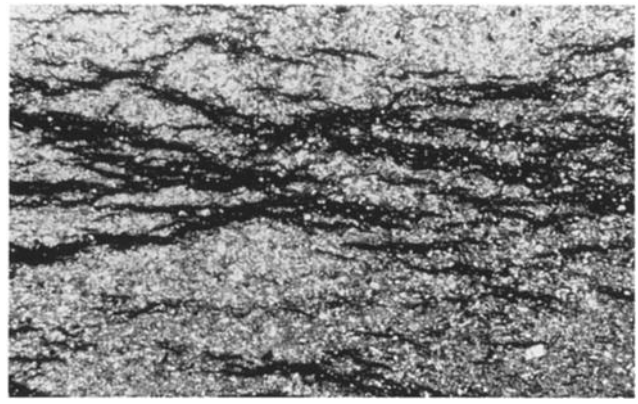


Figure 12. Thin section photomicrograph (plane light) of siltstone red bed showing variations in development of hematite cement. Photomicrograph is 3.5 mm across.

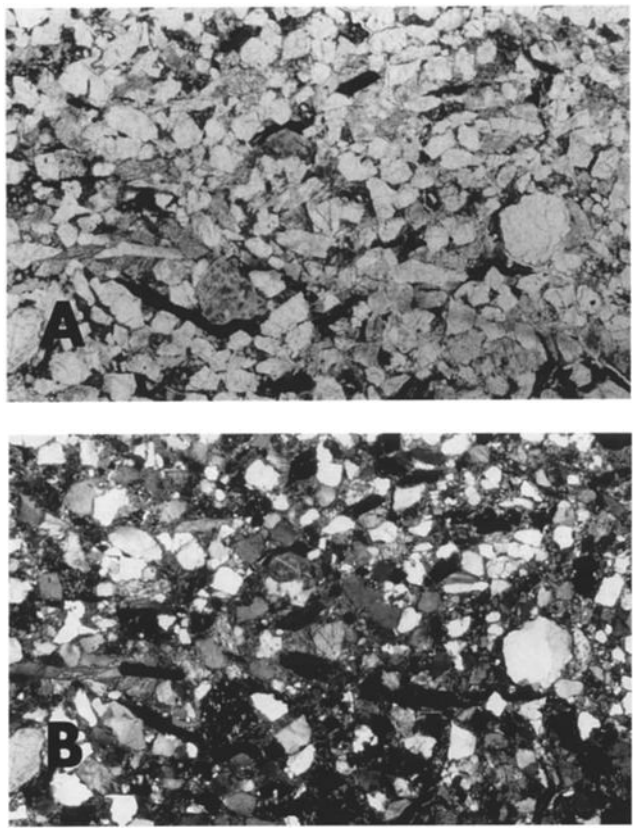


Figure 13. Thin section photomicrograph of pedogenic clay development. Note lack of hematite cement. (a) Plane light. (b) Polarized light. Photomicrographs are 3.5 mm across.

few samples. Thus the characteristic remanance is in hematite, which occurs as a chemically precipitated cement.

Red beds can form in a variety of environments, including desert, aeolian, delta plain, and alluvial settings [*Turner*, 1980]; the main controlling factor seems to be the presence of an oxidizing geochemical environment. They can be classified as either chemical or detrital, depending on how they formed [*Pye*, 1983]. Chemical red beds form in situ by chemical precipitation of iron-bearing phases, while detrital red beds form by erosion, transport, and redeposition of preexisting red sediments. Chemical red beds may form through pedogenetic or diagenetic processes [*Pye*, 1983]. Pedogenetic red beds form in exclusively subaerial environments during the development of distinct weathering profiles, while diagenetic red beds form because of alteration of iron-bearing minerals and flow of groundwaters within the sediments [*Pye*, 1983]. Diagenetic red beds have no connection to surface-weathering processes.

Walker et al. [1981] and *Larson et al.* [1982] studied red beds in the Triassic Moenkopi Formation and concluded on the basis of petrographic and paleomagnetic evidence that the hematite CRM in the red beds had formed over geologically long periods of time. Petrographically, they found that euhedral crystals of hematite occurred as void infillings and as replacement of pyrite and detrital silicate minerals, and hematite was intergrown with several generations of other authigenic minerals. Paleomagnetically, they found both normal- and reversed-polarity components, weak intermediate remanence directions in some samples, and a streaked distribution of remanence directions within one member of the Moenkopi. Thus they proposed a model whereby the authigenic hematite formed from alteration of iron-bearing phases over tens of millions of years.

However, geologic and paleomagnetic evidence in the Pozo red beds suggests that the hematite formed relatively early in the history of the red beds, probably within 10^5 years or less after their deposition. Geologic evidence for early formation is fivefold. First, in the field it can be seen that the red coloration follows bedding but does not appear to be controlled by grain size. This observation argues against groundwater circulation controlling red bed formation, as the contacts between red and nonred units would be random or controlled by lithology, which is not seen in the Pozo red beds. Second, the presence of horizon differentiation, such as the leached and red zones, suggests the development of a weathering profile that would not be found in diagenetic red beds. Long-term diagenetic reddening should affect units on scales of meters to tens of meters or more as groundwater flows through them. Third, no primary bedding structures were found in the red beds, such as laminations or crossbedding. Pedogenesis destroys primary bedding structures, while diagenetic processes often preserve such features [*Pye*, 1983]. Fourth, petrographically there is relatively little detrital magnetite, and the hematite rarely occurs within grain-shaped voids or as pseudomorphs after magnetite (i.e., martite). This suggests that the magnetite was dissolved prior to cementation and possibly prior to deposition. Last, the depositional environment of fluvial overbank deposits is consistent with the early formation of red beds for two reasons: an alluvial plain is an overall oxidizing environment [*Turner*, 1980], and flood plains are inactive, or episodically active, geomorphic environments. This allows minerals to remain at or near the earth's surface for rapid weathering [*Pye*, 1983].

Paleomagnetic evidence for early formation is threefold. First, the normal- and reversed-polarity sites are antipodal (Figure 9), which suggests that overprints have been removed. Second, discrete zones of either normal or reversed polarity can be seen in the Pozo Grade section (Figure 5), which would not be likely to occur if the magnetic signal was an overprint. Long-term development of red beds would not likely produce discrete, antipodal polarity zones but rather smeared data with differing normal- and reversed-polarity directions. Last, the in situ directions are different than present-day directions, so the signal cannot be present-day overprint.

The paleomagnetic and geologic evidence summarized above suggests that the hematite remanence formed within a few thousand to a few hundred thousand years of deposition and not over millions of years. The hematite is interpreted to have formed during and soon after deposition during early pedogenesis above the local (average) water table and not over millions of years because of groundwaters flowing through the rocks long after burial.

The formation of pedogenetic red beds can be described as a two-stage process. First, iron-bearing phases are weathered from the parent material and iron is released [*Pye*, 1983;

Retallack, 1991]. The iron is then precipitated, although the processes by which this happens are not well understood. Ferric iron is precipitated in almost all oxidizing environments; it may be precipitated as ferrihydrite, an unstable iron hydroxide [*Pye*, 1983]. The ferrihydrite is then altered to either hematite or goethite, with the goethite then dehydrating to hematite. The precipitation of ferric iron may occur from groundwaters during diagenetic red bed formation, although *Retallack* [1991] suggests that this may be less common as groundwater is often reducing rather than oxidizing because the oxygen is used by microbes and by the alteration of mafic minerals.

Figure 14 is a schematic illustration of how we envision the hematite to have formed. We envision that the hematite formed within the overbank environment; sediments would have been deposited episodically during flooding, then left subaerially exposed until the next flooding event. Iron would have been released into flowing water by the dissolution of magnetite in the underlying granite, as evidenced by the lack of detrital magnetite in the red beds. The iron may have been transported adsorbed onto clay particles, which would then have been left behind as the flood waters receded. During subaerial exposure and weathering, the sediments would have dried out, and minerals would have precipitated from ions carried in solution. The initial precipitate may have been ferric hydroxide which then later altered to goethite [*Pye*, 1983]; either the ferric hyroxide or the goethite then dehydrated to hematite as the sediments sat above the water table for long periods of time or perhaps during early burial. The common occurrence of goethite as a modern surficial-weathering product but not hematite suggests that hematite may, in fact, develop during earliest burial. The early formation of hematite as a cement may have prevented later groundwater flow from remagnetizing the Pozo red beds thus preserving the primary remanance direction.

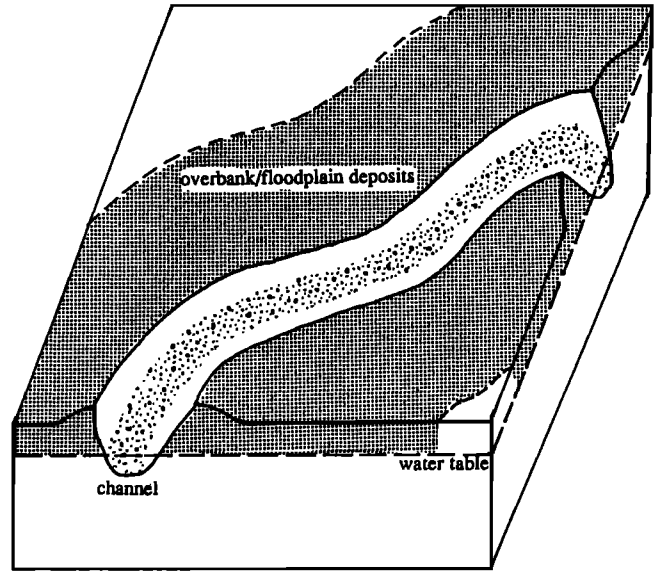


Figure 14. Schematic illustration of hematite formation in an overbank environment.

5. Discussion

When evaluating paleomagnetic data it is critical to interpret a body of data, and not just consider the result of one individual study. Figure 2a is a summary of paleomagnetic results from cratonal rocks in California. The Salinian paleolatitude derived from this study cannot be distinguished from similar results of the Sierra Nevada batholith and clearly shows that Salinia was not allochthonous with respect to rocks of eastern California in the Late Cretaceous. This diagram also illustrates that Sierran paleomagnetic data show consistent offsets of about 6°, which allows for as much as a 500-700 km of northward motion since the Late Cretaceous.

Salinia and the Sur-Obispo terrane have been interpreted to have amalgamated in the Late Cretaceous based on the presence of granitic clasts within a red bed sequence on the Sur-Obispo terrane that borders the Sur-Nacimiento fault [*Vedder et al.*, 1980, 1983]. These clasts were inferred to have been derived from Salinian granites and hence to represent an overlap sequence or tie point. However, three of these granite clasts have been found to have different isotopic compositions and ages from the Salinian batholith (R. Kistler, USGS, personal communication, 1990), making a correlation unlikely. Therefore the timing of amalgamation of Salinia and the Sur-Obispo terrane is unconstrained in the Paleogene.

Thus, when previous workers have developed terrane histories for Salinia [e.g., *Kanter and Debiche*, 1985], they used paleomagnetic data from SO to also reflect the motion history of Salinia. *Champion et al.* [1981, 1984] were the first to arguably work on rocks of Salinia itself. They analyzed deep-sea fan strata of the Late Cretaceous Pigeon Point Formation (PP in Figure 2c) and Paleocene strata of Point San Pedro (PSP in Figure 2c). Although they assigned the Pigeon Point area to Salinia, basement rocks are not exposed for verification. Granites crop out on the Farallon Islands to the west of the locality, but it is not clear that these rocks extend to Pigeon Point, and granitic rocks within the conglomerates do not match granitic rocks in the Santa Lucia Range [*Champion and Kistler*, 1991]. Similarly, the Point San Pedro strata lie in fault contact with (northern block) Salinian granodiorite (at Montara Mountain) on one side and rocks of the Franciscan Complex on the other side. Additionally, based on conglomerate clast provenance studies and an older Campanian age date, *Grove* [1993] concluded that Pigeon Point rocks have a closer affinity to units west of the Nacimiento Fault, that is, SO. It is now our opinion that these studies are all related to SO and that none of them clearly characterize the history of Salinia.

The only paleomagnetic study suitable for paleolatitude estimates that can be clearly associated with Salinia was the study of *Kanter* [1988] on the Eocene Butano Sandstone (BUT, Figure 2a). Her results are consistent with geologic evidence which suggest that Salinia had to be "in place" with respect to eastern California by the Eocene for it contains the Butano Sandstone which is laterally equivalent to the Point of Rocks Sandstone [*Clark and Nilsen*, 1973] located east of the San Andreas Fault. On that basis, we have plotted the Butano (BUT) result with the "cratonal" California results in Figure 2a.

A summary of available paleomagnetic data from cratonal North America, the Sur-Obispo terrane, and the Peninsular Ranges terrane, shown in Figure 3, illustrates the very different relative paleolatitudes of North America, the Sur-Obispo terrane, and the PRT in the Late Cretaceous. The paleomagnetic data from the Sur-Obispo terrane and the PRT consistently show large offsets of 12° to 22° from North America at 80 Ma. These offsets can be distinguished from the limited amount of offset (<6°) suggested by the Salinian and Sierran data. These results suggest that the amount of offset increases away from the craton within an oblique convergent margin. Thus the most outboard terranes, that is, SO which is composed mainly of accretionary prism material, have been translated the farthest, while more inboard terranes, such as Salinia, show decreasing amounts of motion. In this view, even the Sierra Nevada batholith, along with Salinia, has been transported northward 500-700 km as part of distributed dextral shear along the North American cratonal margin since the Late Cretaceous. This model may also apply to other terranes along the western margin of North America. *Tarduno et al.* [1989] and *Hagstrum* [1990], among others, have documented large amounts of transport of the Franciscan Complex in northern California, which is consistent with the relative amount of offset seen in SO, an equivalent-type terrane.

6. Conclusions

The paleolatitude of Salinia in the Late Cretaceous was within 6° of its current location on the margin of cratonal North America (after correction for Neogene structural disruption). These results are derived from an early hematite CRM within Late Cretaceous (75-85 m.y.) red beds deposited on granites of the central block of Salinia. Thus the terrane history of Salinia can be separated from the history of the Sur-Obispo terrane, which has paleomagnetic results that may indicate significantly lower paleolatitudes during the Late Cretaceous. These results lead to a revised model for the geologic history of coastal California, where SO and the PRT originated farther to the south within the same oblique subduction zone and were transported northward. Salinia originated within 500-700 km of its present position and only amalgamated with SO sometime in the late Paleogene. This model suggests that within an oblique subduction zone offset of terranes is partitioned across the margin with the greatest amount of transport occurring outboard and decreasing amounts of offset occurring toward the craton center.

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